SHS Lab Test — 2026-05-28

Clark South Ballast Tank, WHOI

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2026-05-28 (data through 2026-05-29 ˜11:10 UTC)

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1 Setup

Six dunk cycles in the Clark South saltwater ballast tank (30–32 PSU) at WHOI. A fan provided steady airflow to the WXT520 head throughout.



Figure 1: Clark South ballast tank. Fan and WXT520 visible at left.

Instrument	ID	Rate
RBRduo3 T.T (dry/wet bulb)	SN 233860	3 s
Vaisala WXT520 (met)	ASWXT203	1 min
Seabird GPCTD (tank reference)	—	1 Hz

Event log (UTC, 2026-05-28)

Cycle	IN	OUT	Duration
IN-1	13:11	13:27	16 min
IN-2	13:56	14:01	5 min
IN-3	14:56	15:01	5 min
IN-4	15:42	15:49	7 min
IN-5	16:30	16:34	4 min
IN-6	18:30	18:50	20 min

Instruments on at 13:08; recovered 2026-05-29 ˜11:10 UTC.

Ambient conditions (WXT, 13:08–19:00 UTC)

Variable	Mean	Std
Air temperature (°C)	18.17	0.15
Relative humidity (%)	64.4	0.8
Barometric pressure (mbar)	1006.89	0.12
Wind speed (scalar mean, m/s)	1.81	—

2 Humidity: WXT520 vs. psychrometric

- **During active test (13:08–18:50 UTC):** The two methods agree well at ∼64 % RH in air, with short spikes during each dunk as the wet bulb equilibrates with the saltwater.
- **Overnight (after IN-6):** The psychrometric RH rises to ∼100 % because the wick remained wet after the last dunk and the sensor was left in air overnight. WXT520 stays stable at ∼65–68 %. This is an artefact of the test procedure, not a sensor fault.
- **Residual offset:** Once the wick is saturated, psychrometric RH overshoots WXT by ∼4 pct-pt ($\Delta e \approx +0.8$ hPa) – likely a small wick-induced evaporative enhancement. To be monitored in future tests.
- **Action item:** A live WXT display during the test would give immediate feedback on wick state (per Simon's suggestion).

3 Temperature and sensor response

SHS vs. in-tank GPCTD reference

T_{wet} approaches tank temperature during each dunk; T_{dry} stays near ambient due to its larger thermal mass. Compass and tilt are stable throughout, confirming no sensor movement.

Response-time fits (OUT events)

Method. An exponential model $y(t) = A + B\,e^{-t/\tau}$ is fitted by nonlinear least squares to T_{dry}, T_{wet}, depression ($T_d - T_w$), and psychrometric RH over a window from each OUT time to min(OUT + 25 min, next IN). τ is the e-folding recovery time.

OUT	τ_{T_d} (s)	τ_{T_w} (s)	τ_{dep} (s)	τ_{RH} (s)
13:27	389	51	58	56
14:01	1475*	29	31	30
15:01	74	61	65	64
15:49	130	42	51	48
16:34	139	41	54	51
18:50	67	27	35	33
Median	134	42	53	50

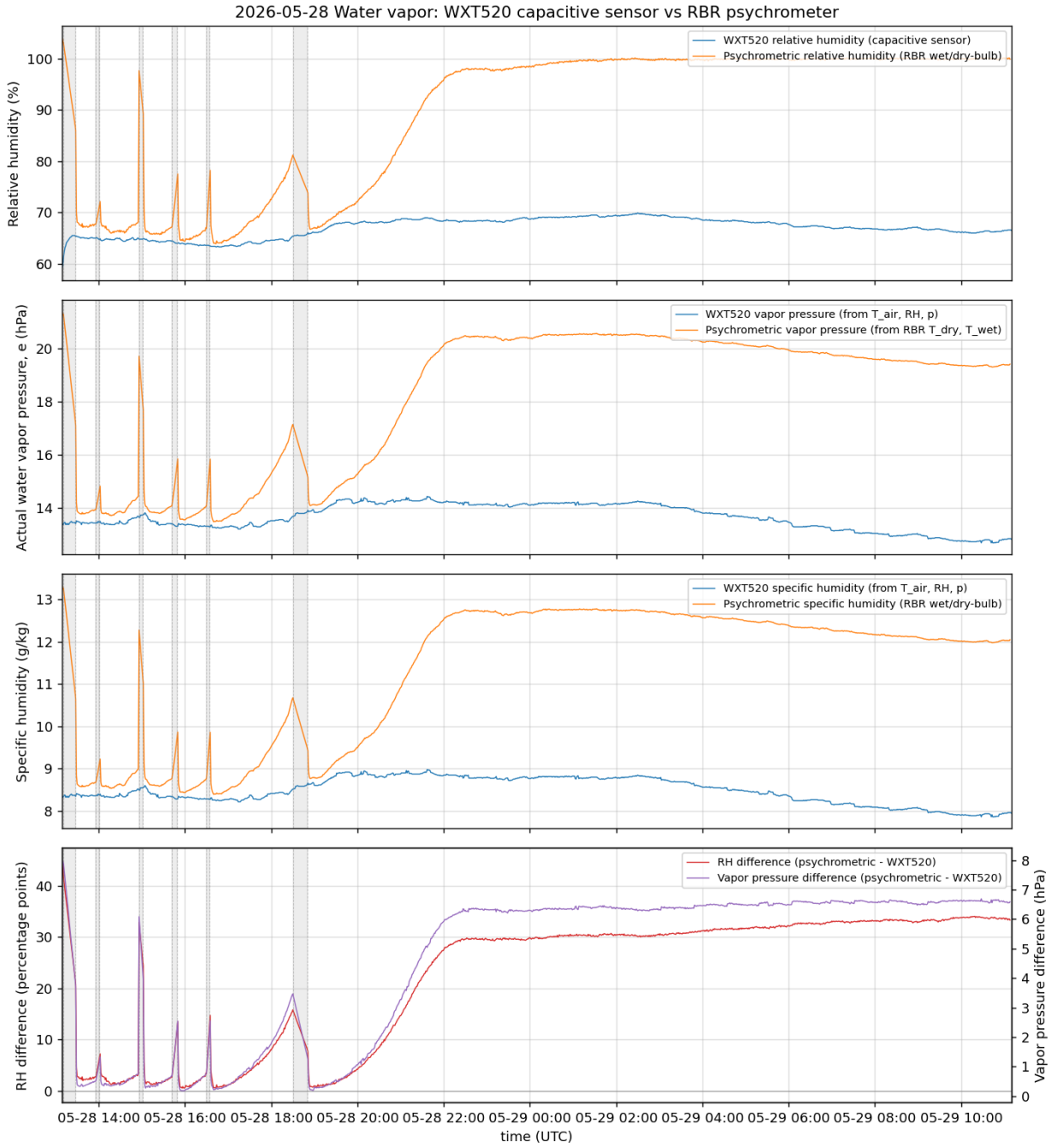


Figure 2: Head-to-head comparison of the WXT520 capacitive humidity sensor against the RBR psychrometric calculation (wet/dry-bulb). Grey bands mark the six in-water periods. **Top:** relative humidity. **Middle two:** actual vapor pressure and specific humidity. **Bottom:** signed differences (psychrometric − WXT520).

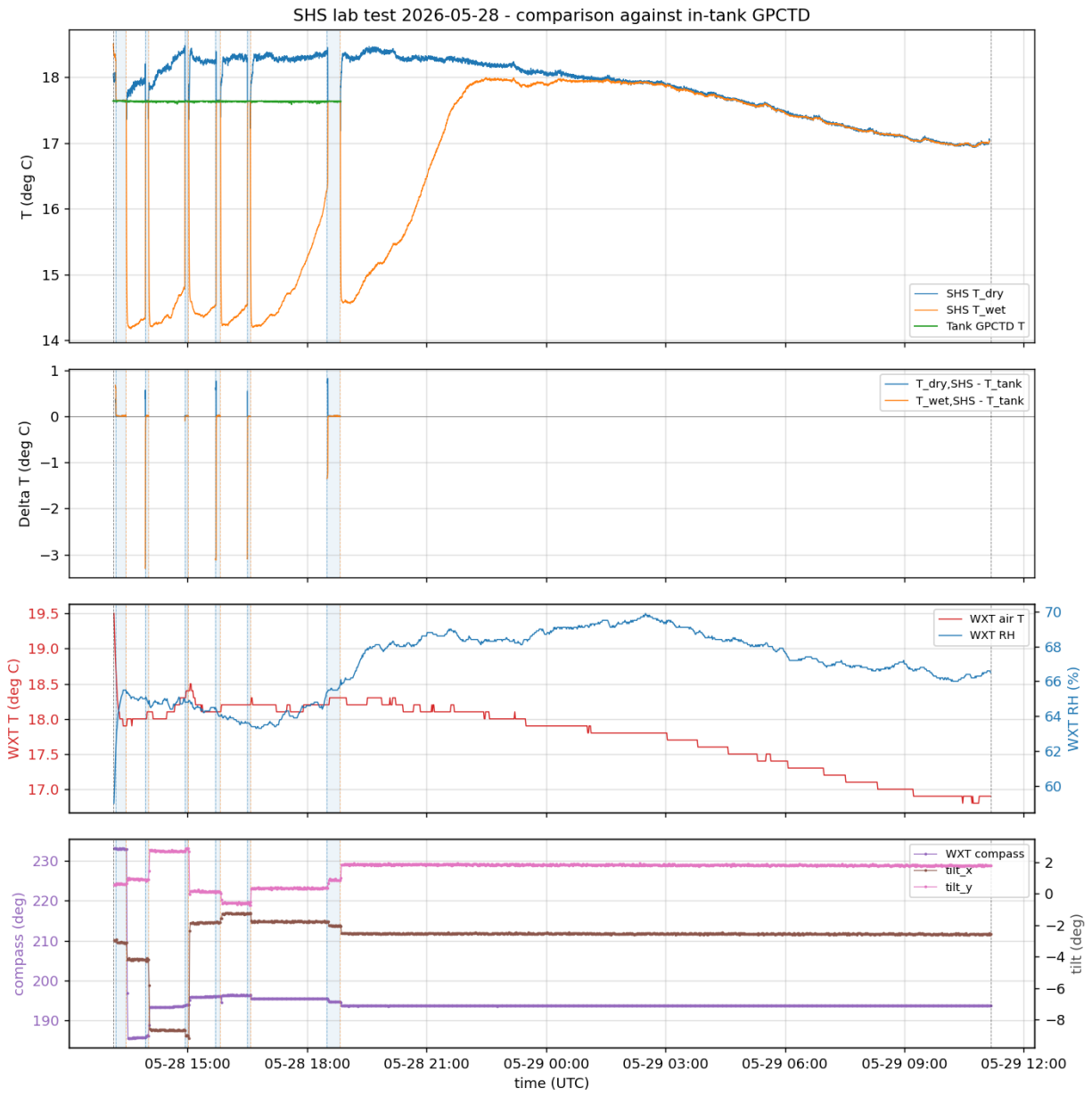


Figure 3: SHS dry/wet-bulb temperatures and GPCTD tank reference. **Top:** absolute temperatures. **Second:** ΔT (SHS − tank), in-water periods only. **Third:** WXT air temperature and RH. **Bottom:** WXT compass and tilt (sensor attitude diagnostic). Blue bands: in-water periods.

* window capped at 25 min; fit unreliable.

T_{wet}, depression, and RH recover in ∼30–65 s. T_{dry} is the slow component (median $\tau \approx 134$ s); psychrometric RH inherits this when computed through T_{dry}.

Wet-bulb cooling (IN events)

T_{wet} peaks in cooling rate at ∼20–50 s after submersion and plateaus near tank temperature within ∼1 min.

4 Wind

Mean wind speed ≈1.8 m/s (fan); compass stable at ∼195°.

Note on `wspd_min`/`wspd_max`: These fields originate from the WXT520's native serial output fields `Sn` (speed minimum) and `Sx` (speed maximum). The ASIMET GILWND45 front-end board passes them through unchanged via a direct `aR1` relay — it applies no accumulation logic of its own (`https://uop.whoi.edu/UOPinstruments/frodo/asimet/wxt520ifcmnd.html`). The exact averaging window used by the WXT520 internally for `Sn`/`Sx` versus `Sm` is configurable and documented only in the Vaisala WXT520 user manual, which has not yet been consulted. Because `wspd_min` exceeds `mean(wspd11)` in 99 % of records, the two quantities are clearly not from the same population. All reported wind statistics in this report use `wspd11` directly.

5 Next steps

1. **Consult Vaisala WXT520 user manual.** Determine the exact accumulation window for `Sn`/`Sx` (minimum/maximum wind speed) vs. `Sm` (mean), and confirm whether the ASIMET deployment uses the default WXT520 configuration. This will fully resolve the `wspd_min` > `mean(wspd11)` discrepancy.
2. **Pre-wet the wick before each test.** IN-1 through IN-3 showed undersaturation. Soak the wick in fresh water for ∼5 min before the start of each test session.
3. **Add a live WXT display during tests.** A real-time readout of WXT RH and air temperature would give immediate feedback on wick saturation and ambient stability (per Simon's suggestion).
4. **Repeat the overnight soak intentionally.** The post-IN-6 overnight data showed psychrometric RH rising to ∼100 % as the wet wick dried in air. A controlled version of this (sensor left in known-humidity air overnight) would provide a cleaner characterisation of wick drying rate and long-term psychrometric drift.
5. **Quantify the ∼4 pct-pt psychrometric offset.** Once wick saturation is reliable (IN-4 onward), the psychrometric RH overshoots WXT by ∼4 pct-pt. Repeat tests with a calibrated reference hygrometer to determine whether this is a systematic wick bias, a WXT calibration offset, or a pressure/temperature correction issue.

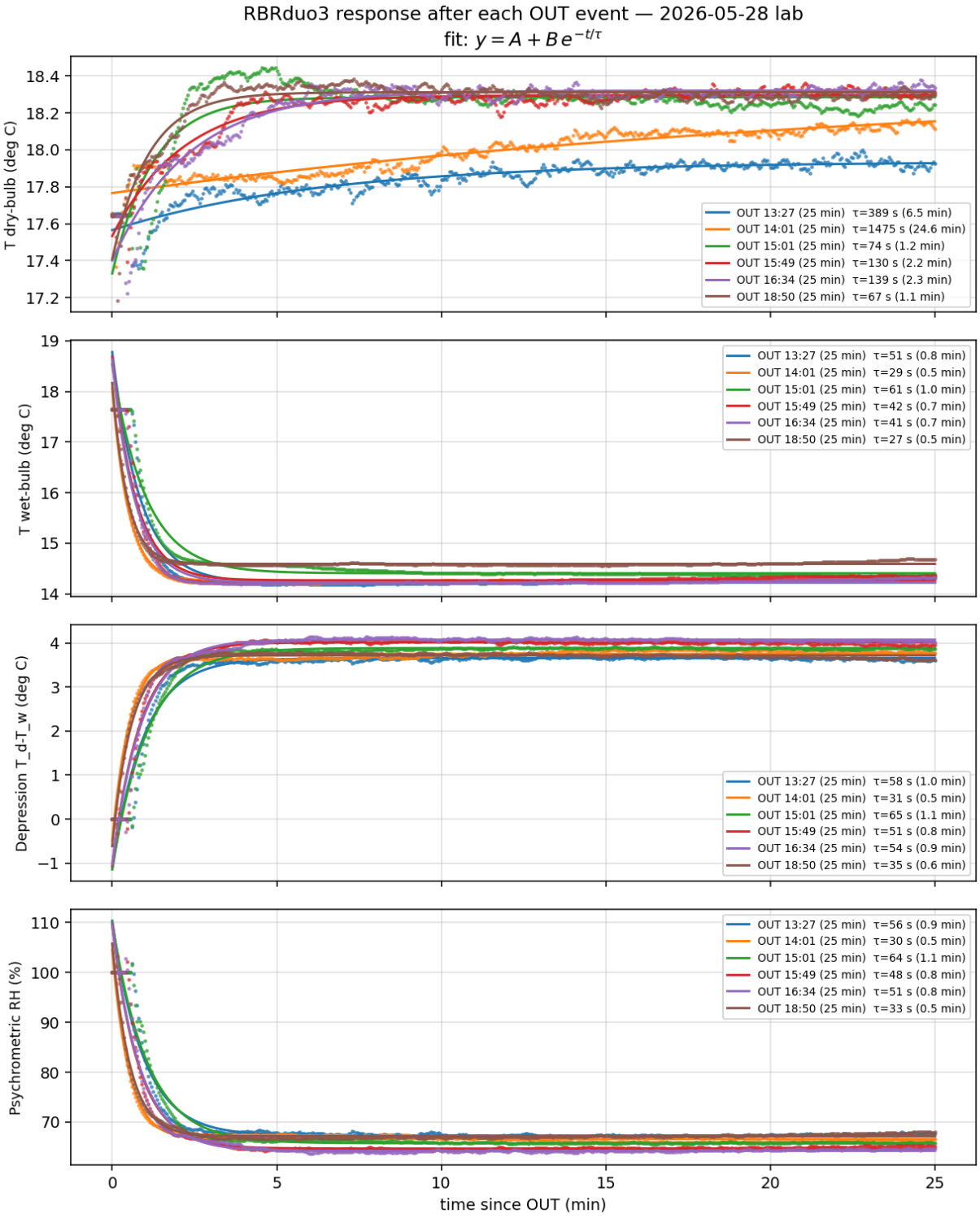


Figure 4: Exponential recovery fits for all six OUT events.

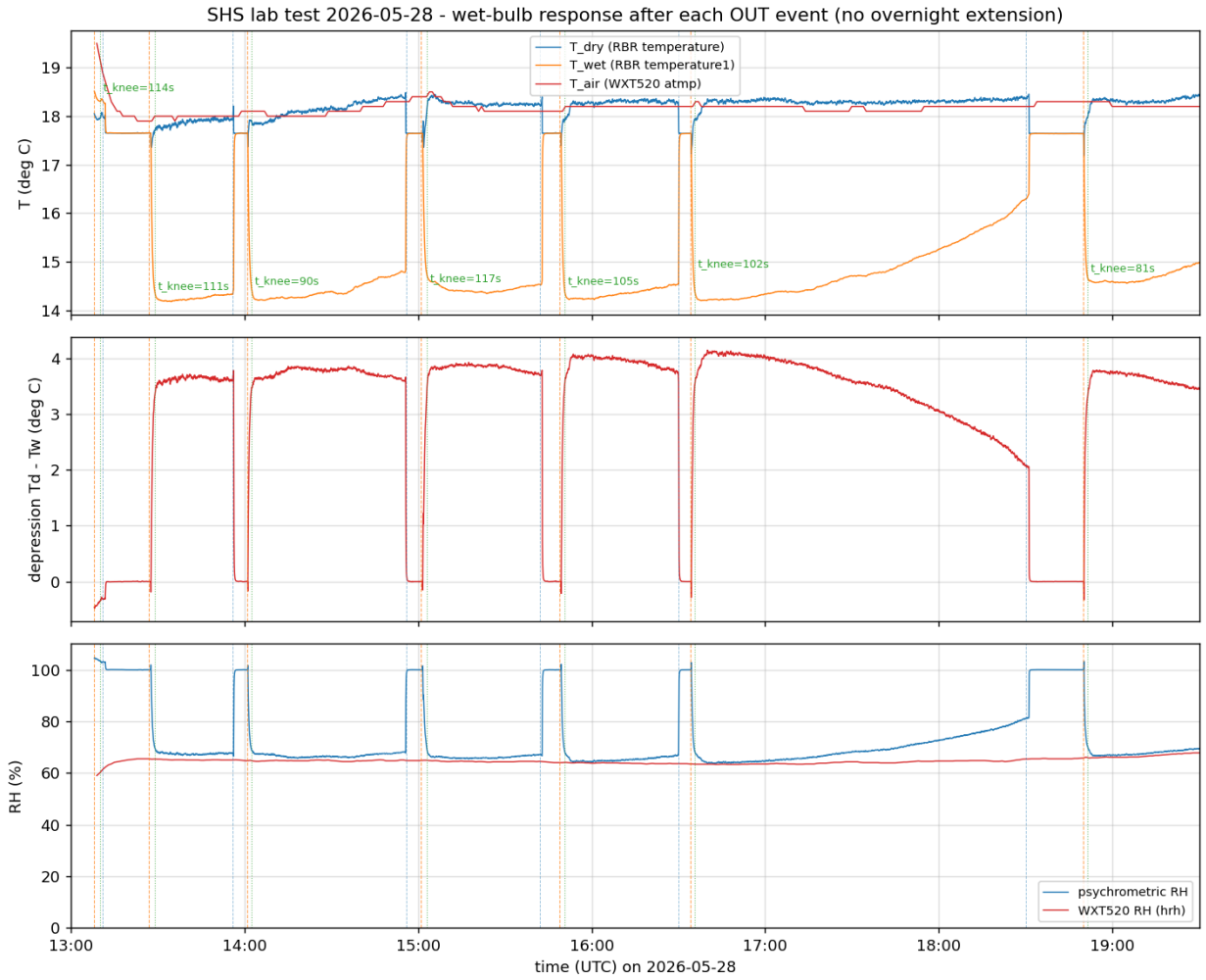


Figure 5: Wet-bulb cooling for each IN event. Orange dashed: OUT time.

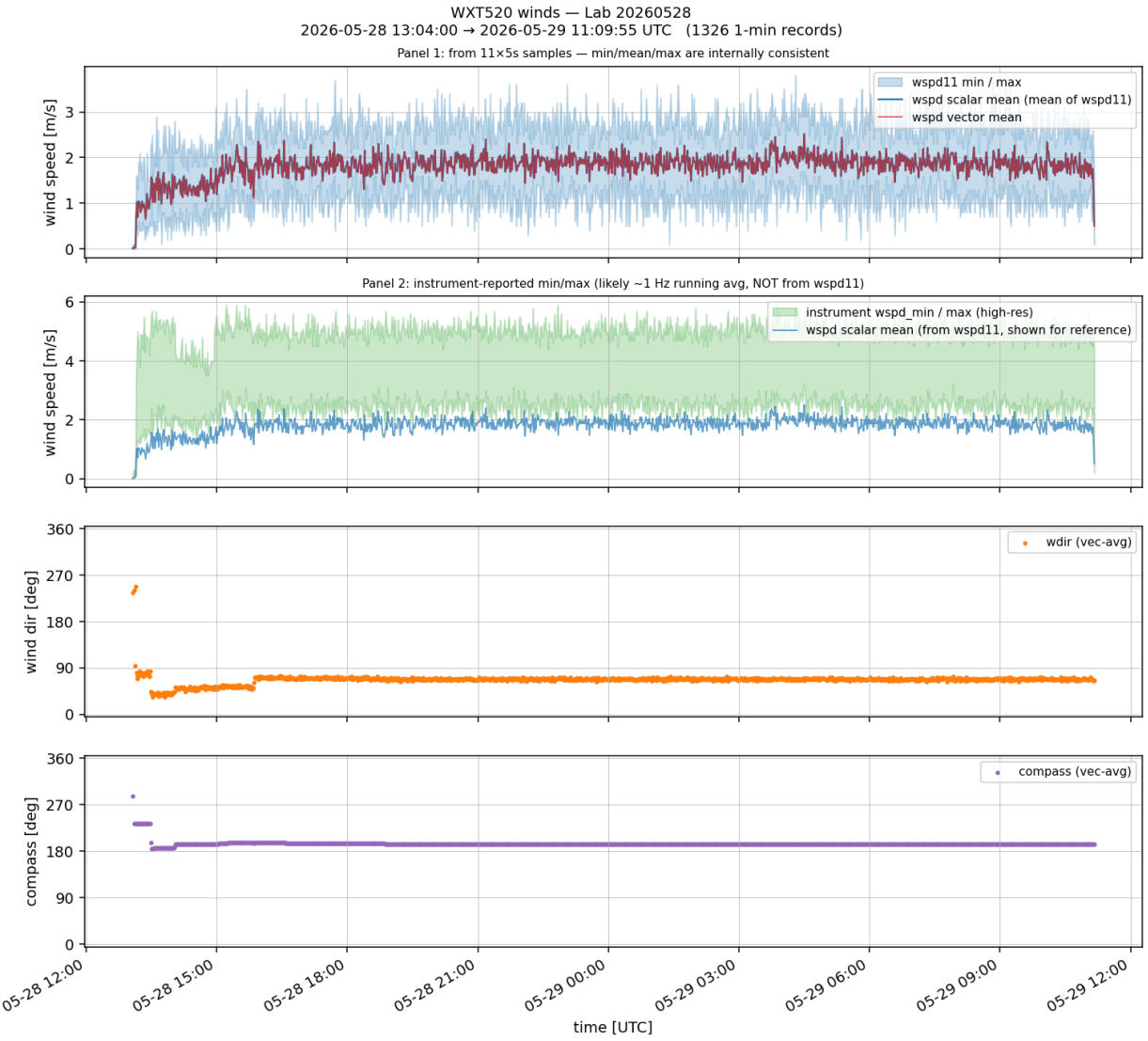


Figure 6: WXT520 wind over full deployment. **Top:** shaded band = min/max of `wspd11` (11 five-second samples/min); blue = scalar mean; red = vector mean. **Middle:** instrument-reported `wspd_min`/`wspd_max` (different population — see text). **Bottom two:** wind direction and compass.